5. (a) The circular gate ABC has 1m radius and is hinged at B. Neglecting atmospheric pressure, determine the force P just sufficient to keep the gate from opening when h= 12m. (8)

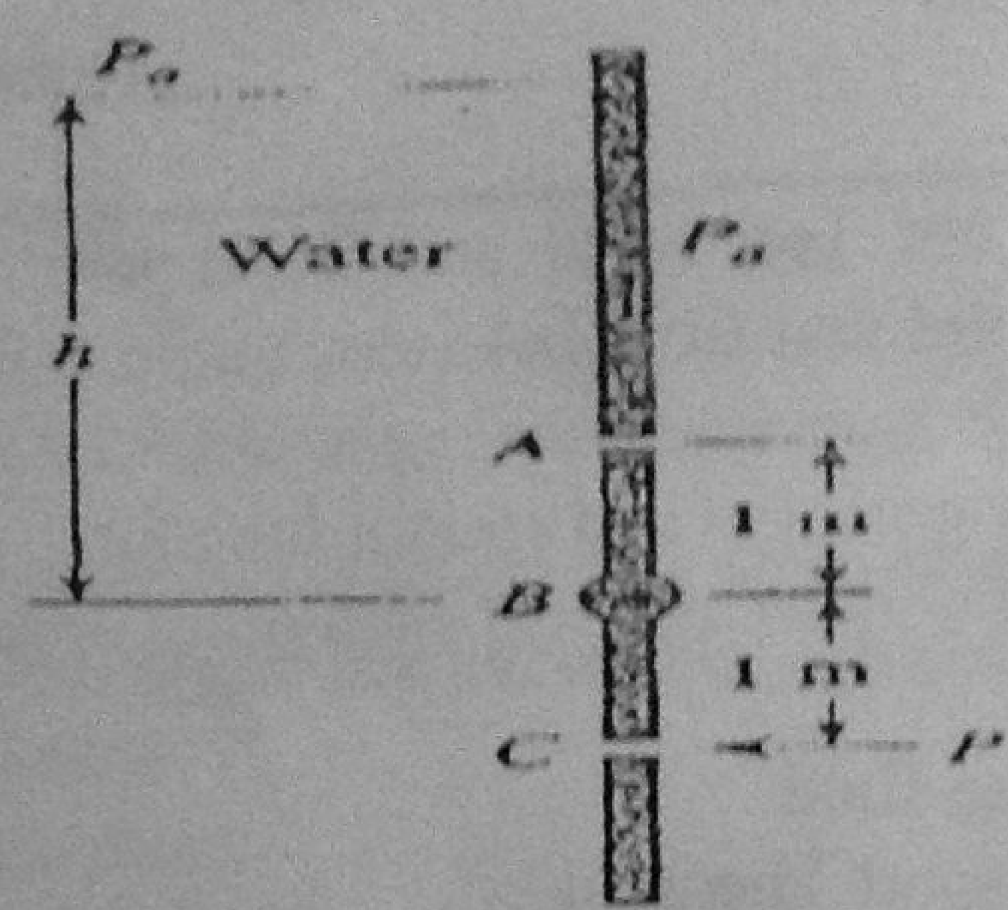

Figure Q5

(b) Consider a flow with velocity component u=0, $v = y(y^2 - 3z^2)$ and $w = z(z^2 - 3y^2)$.

i. Is this is one, two or three dimensional flow?

ii. Demonstrate whether this is in an incompressible or compressible flow?

iii. Derive a stream function for this flow. (1 + 2 + 4= 7)

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE

I SEMESTER 2012-2013

CE F212 Transport Phenomena Dated: 4.10.2012

Mid Semester Test (Closed Book)

Max. Marks: 70 Duration: 90 minutes

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1. (a) A number of common substances are Tar, Sand and Toothpaste. Some of these material exhibit characteristic of both solid and fluid behavior under different condition. Explain and give examples.
   (b) Derive Laplace equations for velocity potential and stream function.
   (c) Prove that streamlines and equipotential lines intersect each other orthogonally.
   (d) Define the metacentric height.
   (e) Define Newtonian and Non-Newtonian fluid.
   (f) Write the equation for Reynolds Transport Theorem.
   (g) Differentiate between control volume and system approach. (2 x 7= 14)

2. A conical pointed shaft turns in a conical bearing. The gap between shaft and bearing is filled with heavy oil having the viscosity 0.2 Ns/m^2. Obtain an algebraic expression for the shear stress that acts on the surface of the conical shaft. Calculate the viscous torque that acts on the shaft.

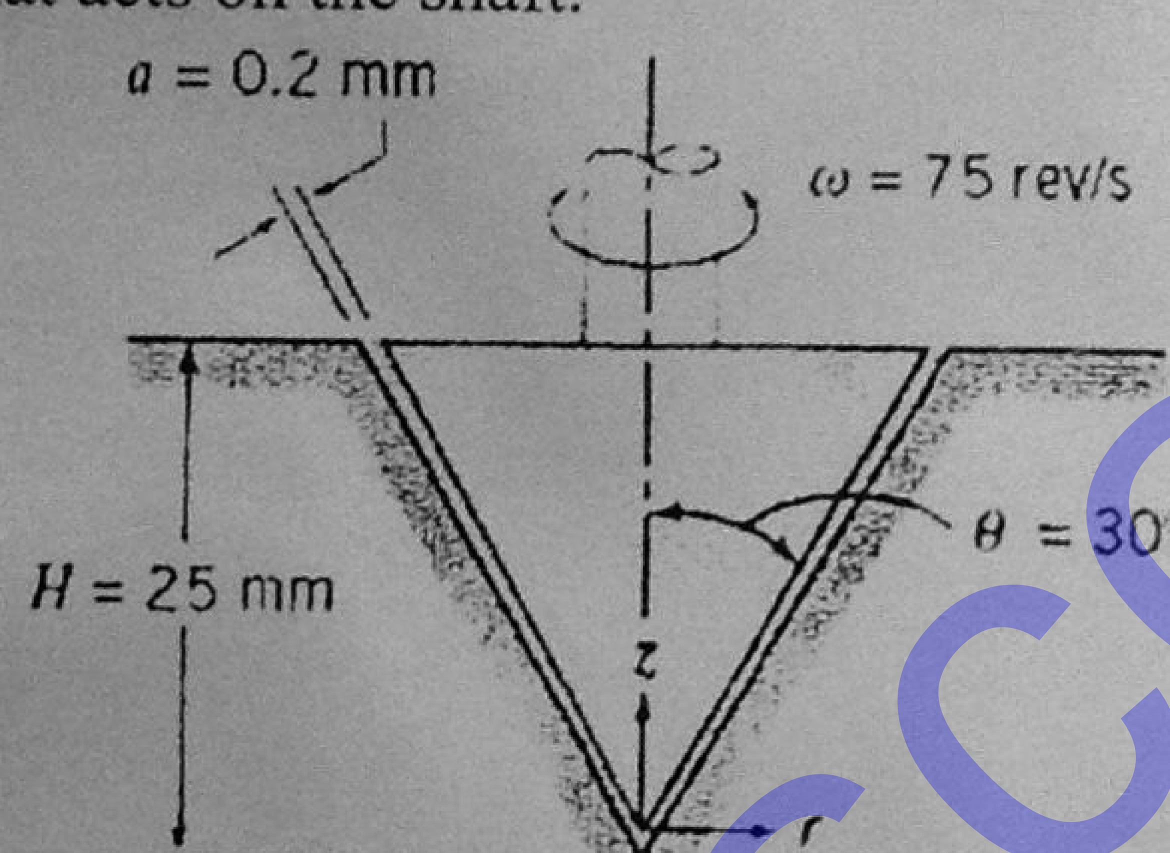

Figure Q-2 (10 +10 = 20)

3. Compressed air exhausts from a small hole in a rigid spherical tank at the mass flow rate of $\dot{m}$, which is proportional to the density ρ of the tank. If ρ_0 is the initial density in the tank of volume, V, derive an equation for the density changes as function of time after the hole is opened. (15)

4. A rectangular block of width b and a submerged depth of H has its centre of gravity at the waterline. Find the metacentric height in terms of b/H and derive an expression for stable equilibrium in terms of b/H. (Properties in the tank are uniform but time dependent). (3+3 =6)

P.T.O.